

No. of Functional Features Included in the Solution

Date	29 June 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Interactive Prediction Form	Responsive web form gathering 11 applicant parameters (income, credit history, dependents, etc.) for eligibility evaluation	Frontend: predict.html / script.js	Done	1.0 Mark
2	XGBoost Prediction Model	Machine Learning model (XGBClassifier) evaluating applicant risk profile and predicting approval status (accuracy: ~82%).	Backend: app.py / rdf.pkl	Done	1.5 Marks
3	Two-Tier Input Validation	Form validations implemented on browser (Bootstrap constraint feedback) and server (positive bounds and categories mapping)	Frontend / Backend: script.js / app.py	Done	1.0 Mark
4	Live Loan Burden Calculator	Computes and displays the estimated annual loan-to-income payment burden ratio in real-time as users modify income/loan amounts.	Frontend: script.js (updateBurden)	Done	0.5 Mark
5	Draft Persistence Control	Local browser storage hooks allowing users to Save Draft or Load Draft of form entries across multiple web sessions.	Frontend: script.js (localStorage)	Done	0.5 Mark
6	Dual-Theme Interface	Creative banking theme toggle (Light / Dark mode) with device layout optimization and CSS variables variables-driven design.	Frontend: style.css / script.js	Done	0.5 Mark
7	Database Application Storage	Relational database to log loan applicants' personal profiles, prediction history, and eligibility outcomes.	Database: <i>SQLite</i> / <i>PostgreSQL</i>	Pending	0.0 Marks (Planned)
8	User Auth & Admin Control	Login and registration module enabling secure portal access for applicants and role-based views for bank loan officers.	Security: <i>Flask-Login</i> / <i>JWT</i>	Pending	0.0 Marks (Planned)

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	6
Core / Must-Have Features	5 (Form, ML Prediction, Validation, Database, Auth)
Additional / Nice-to-Have Features	3 (Burden Calculator, Draft Persistence, Dual-Theme UI)
Features Tested & Verified	6

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Responsive Dual-Theme Layout, Form Completion Progress Bar, Quick Scenario Presets.
2	Backend / Logic	2	Flask web routing endpoint handlers, Server-side categorical validation checks.
3	Database / Storage	1	Client-side LocalStorage draft persistence (<i>Pending: server-side relational database history</i>).
4	API / Integration	1	Pickle serialization interface to load model features scaler and XGBoost classifier.
5	Security / Authentication	1	Client-side input boundary constraints (<i>Pending: backend secure login/session manager</i>).